

MECH 191 — Metallurgy

Week 14

Welding Processes

November 23, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Arc Welding Processes

SMAW, GMAW, GTAW, and FCAW — how each works, what it's used for, and its quality risks

02

Filler Metals & Thermal Control

AWS filler classification, matching vs. overmatching, preheat, interpass temperature, and carbon equivalent

03

Weld Symbols & Technician Responsibilities

Reading the reference line, identifying weld type and size, and applying the symbol to inspection

Module 13 · Week 14

Welding Processes

What to do this week:

Map FZ / HAZ / BM zones at 50×. Sketch zone boundaries and estimate HAZ width.

Steps:

- 1 Observe — complete feature table
- 2 Sketch — label magnification & specimen
- 3 Answer — 4 short questions

Specimens to Check Out

- Specimen J — carbon steel MIG weld cross-section

How to check out the kit:

- 1 See instructor after class today
- 2 Sign out the specimen tray
- 3 Use the EXAMET-5 at your convenience
- 4 Return kit before leaving class
- 5 Submit completed module with your portfolio

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 ← here Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 · FINAL EXAM

Week 14 — At a Glance

Topics

Welding fundamentals — heat sources, joint types, groove preparations, weld positions, and heat input formula

SMAW — stick welding, coated electrodes, E7018 classification, slag protection, and low-hydrogen storage

GMAW — MIG welding, continuous wire, shielding gases, transfer modes (short circuit, spray, pulsed)

GTAW — TIG welding, non-consumable tungsten, AC for aluminum, DCEN for steel, highest quality

FCAW — flux-cored wire, self-shielded vs. gas-shielded, high deposition rate, structural steel

Preheat & interpass — carbon equivalent, cooling rate control, HIC prevention, verification methods

Weld symbols — reference line anatomy, arrow side vs. other side, weld type, size, length, and tail

Resources

Reference Material

Kalpajian Ch. 30 · Moniz Ch. 27
Fusion Welding Processes & Welding Metallurgy

Visualizers

SMAW — Arc & AWS Classification
GMAW — Transfer Modes & Shielding Gas
FCAW — Self-Shielded vs Gas-Shielded
GTAW — Tungsten & Filler Classification

Key Reference

AWS D1.1 Structural Welding Code
Weld Symbols — AWS A2.4
Preheat Calculator — Lincoln Electric

Next: Week 15 — Weld Metallurgy & Failure

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1

What is welding and how does it create a metallurgical bond?

2

What are the main arc welding processes and when is each used?

3

How do filler metals work and how are they selected?

4

What are preheat and interpass temperature and why are they quality-critical?

5

How do you read a basic weld symbol?

Welding Fundamentals — Joint Design & Heat Input

Joint Types & Groove Preparations

Five basic joint types:

- Butt** — two pieces edge to edge (most common in pipe and plate)
- T-joint** — one piece perpendicular to another (fillet or groove weld)
- Lap** — two pieces overlapping (fillet each side)
- Corner** — two pieces at right angles at their edges
- Edge** — two pieces parallel, welded along their edges

Groove preparations:

Square groove

Thin material; full penetration without bevel; simple and fast.

V-groove

Most common; single or double; bevel angle controls penetration and weld volume.

U-groove

Thick material; reduced weld volume vs. V-groove; more prep cost but less filler.

Bevel groove

Single-sided access only; one plate bevelled, the other left square.

Positions, Heat Input & Bond Formation

How welding creates a metallurgical bond:

A heat source melts the base metal and filler to form a weld pool — as it solidifies, a continuous metallic connection forms across the joint. The joint is not glued; it is a continuation of the metal itself. Microstructure and mechanical properties are determined by the thermal cycle experienced.

Weld positions:

Flat (1G/1F) · Horizontal (2G/2F) · Vertical (3G/3F) · Overhead (4G/4F)
Position affects heat input control, weld pool behavior, and the skill required. Procedure and welder qualifications are position-specific.

Heat Input

$$HI = (V \times A \times 60) / (1000 \times \text{travel speed mm/min}) \rightarrow \text{kJ/mm}$$

Higher heat input → slower cooling → coarser grain structure, wider HAZ, more distortion

Lower heat input → faster cooling → finer grain structure, greater risk of hydrogen cracking in hardenable steels

Heat input specified in the WPS and must be controlled within limits.

SMAW & GMAW — Stick and MIG Welding

SMAW — Shielded Metal Arc Welding

How it works

Consumable coated electrode; the arc melts both the core wire and the flux coating. Molten flux forms a shielding gas and a slag layer that floats on the pool and slows cooling — slag removed between passes.

Equipment

AC or DC power source, electrode holder, work clamp, cables — minimal equipment, highly portable. Can be used anywhere with a power supply.

E7018 — AWS Classification

E = electrode · 70 = 70 ksi (480 MPa) minimum tensile · 1 = all positions · 8 = low-hydrogen flux, DC or AC. Must be stored in a heated oven; use within 4 hours of removal.

Advantages / Limitations

Versatile, portable, all-position, tolerant of surface contamination and wind. Lower deposition rate than GMAW/FCAW; frequent electrode changes; slag removal required; skill-dependent.

GMAW — Gas Metal Arc Welding (MIG)

How it works

Continuous bare wire fed from a spool through the gun — no electrode changes. Shielding gas protects the arc. Arc length is self-regulating with a constant-voltage (CV) power source.

Shielding gases

Pure argon: non-ferrous metals (Al, Cu, Ti) — stable arc, good wetting.

Pure CO₂: carbon steel — deep penetration, more spatter, low cost.

75Ar/25CO₂: most common for structural steel — best balance of stability, penetration, and spatter.

Transfer modes

Short circuit: low V/A — thin material and out-of-position; risk of lack of fusion.

Spray: high V/A — flat/horizontal only; high deposition, excellent fusion.

Pulsed spray: combines high rate with out-of-position capability.

Advantages / Limitations

High deposition rate, no slag removal, suitable for automation, good for thin to medium material. Sensitive to drafts (outdoor use needs windshields); requires clean base metal; less portable than SMAW.

GTAW & FCAW — TIG and Flux-Cored Welding

GTAW — Gas Tungsten Arc Welding (TIG)

How it works

Non-consumable tungsten electrode — only the base metal and separately fed filler rod melt. Complete independent control of heat input and filler addition. No spatter; no slag.

AC vs. DCEN

AC for aluminum: positive half-cycle cleans the oxide layer (Al_2O_3 melts at 2050°C , base metal at 660°C); negative half-cycle provides penetration. DCEN for steel and stainless: maximum heat into the workpiece, narrow deep penetration profile.

Applications

Root passes in pipe and pressure vessel welds; thin sheet; stainless steel, aluminum, titanium; any application where quality takes precedence over productivity.

Advantages / Limitations

Highest quality welds; precise heat control; all materials and positions. Slowest arc process; highest skill required; filler by hand limits deposition rate; not economical for thick material.

FCAW — Flux-Cored Arc Welding

How it works

Tubular wire filled with flux fed continuously — same equipment as GMAW but combines continuous wire feeding with slag protection. Higher current density than solid wire gives higher deposition rate.

FCAW-S vs. FCAW-G

Self-shielded (FCAW-S): flux core generates its own shielding — no external gas; highly portable and tolerant of wind; AWS D1.1 approved for structural steel site welding.

Gas-shielded (FCAW-G): external CO_2 /mixed gas added — better mechanical properties and lower fumes; preferred for shop fabrication.

Applications

Heavy structural fabrication, shipbuilding, heavy equipment manufacture — wherever high deposition rate and tolerance to mill scale are required. Out-of-position with appropriate wire.

Advantages / Limitations

High deposition rate; all-position capability; tolerant of mill scale and contamination. Slag removal required between passes; higher fume generation; FCAW-S generally lower toughness weld metal.

Filler Metals — Classification and Selection

How Filler Metals Work

Filler metal melts into the weld pool and becomes part of the deposited weld metal — its composition determines the as-welded microstructure, mechanical properties, and corrosion resistance.

Match or overmatch

Filler selected to meet or exceed base metal minimum tensile strength — an undermatching weld is the weakest point of the joint and will fail first.

Chemical compatibility

Filler must be chemically compatible with the base metal and process — different processes use different filler forms (coated electrode, bare wire, flux-cored wire, rod).

Dilution

Deposited weld metal is a mix of filler and melted base metal. SMAW: 20–30% dilution. GMAW: 25–35%. SAW: 50–70%. The WPS specifies the filler to account for dilution.

AWS Classification System

Prefix letter identifies the form: E = electrode; ER = electrode or rod; R = rod only.

E7018 — SMAW low-hydrogen electrode

E = electrode (SMAW consumable coated rod)

70 = 70 ksi (480 MPa) minimum tensile strength

1 = all-position capability (1G, 2G, 3G, 4G)

8 = low-hydrogen potassium flux; DC or AC; bake before use

ER308L — GTAW/GMAW stainless steel wire

ER = electrode or rod

308 = austenitic stainless filler for 304/308 base metal

L = low carbon ($\leq 0.03\%$ C) — prevents sensitization

ER4043 — GTAW/GMAW aluminum wire

ER = electrode or rod

4043 = aluminum-silicon filler for 6xxx series alloys

Lower cracking risk than 5xxx fillers for heat-treatable alloys

Preheat & Interpass Temperature — Controlling the HAZ

Carbon Equivalent & Preheat

Preheat is the minimum temperature the base metal must reach before welding begins. It slows the cooling rate of the weld and HAZ — preventing brittle martensite formation and allowing hydrogen to diffuse out before it can cause cracking.

Carbon Equivalent (CE) Formula:

$$CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$$

Higher CE = higher hardenability = higher preheat required.

CE < 0.40

Generally no preheat required for thin sections.

CE 0.40–0.60

Preheat typically 100–200°C; required for thicker sections.

CE > 0.60

Preheat 200–300°C or higher; mandatory for most applications.

Interpass Temperature, Verification & PWHT

Interpass temperature minimum

Must not cool below the preheat temperature between passes — critical for multi-pass welds on thick sections where the joint can cool rapidly.

Interpass temperature maximum

Carbon steel typically 250–300°C. Austenitic stainless typically 150°C maximum — excessive heat causes sensitization ($Cr_{23}C_6$ precipitation at grain boundaries, reducing corrosion resistance).

Verification

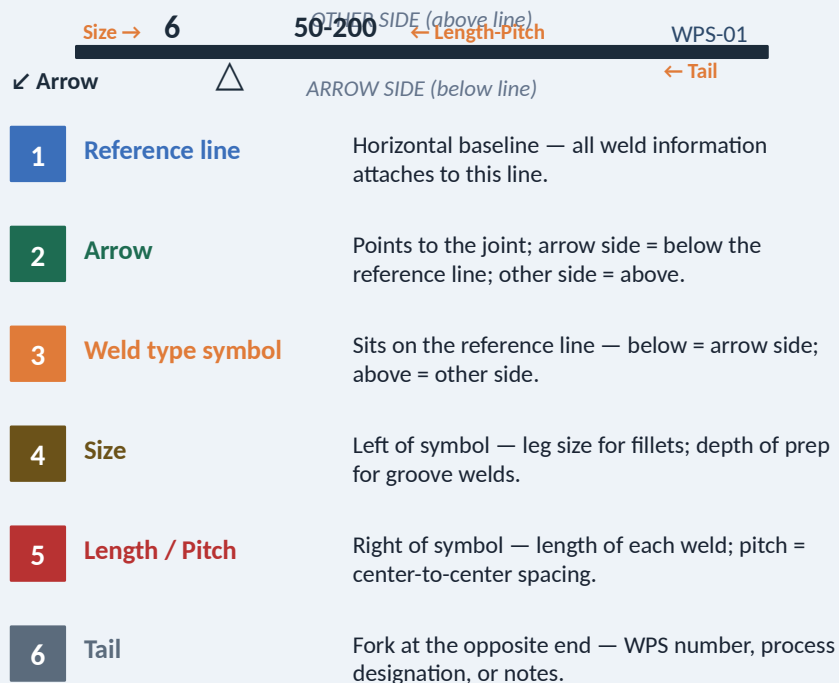
Tempilstik (temperature-indicating crayons), contact thermometers, or infrared pyrometers. Per AWS D1.1: verify at 75 mm from the joint each side, immediately before welding.

Post-Weld Heat Treatment (PWHT)

Stress relief PWHT reduces residual stress and tempers HAZ martensite — mandatory in many pressure vessel and piping codes above certain thickness and CE thresholds. Temperature, heating/cooling rate, and hold time are specified in the applicable code and WPS.

Weld Symbols — Reading the Drawing Standard

Anatomy of a Weld Symbol



Weld Type Symbols & Special Modifiers

-  **Fillet**
T-joints and lap joints; most common weld type in structural fabrication.
-  **V-groove**
Full-penetration butt weld; single V most common for plate; double V for thick sections.
-  **Square groove**
Full penetration in thin material without bevel preparation.
-  **Bevel groove**
One-sided access; one plate bevelled, the other left square.
-  **U-groove**
Thick material — less weld volume than V-groove; more expensive prep.
-  **Plug / Slot**
Weld through a hole or slot in the top piece into the piece beneath.

○ Weld all around • ▸ Field weld • G / M / C Finish (grind/machine/chip) ·
Contour: flat / convex / concave

Technician's Perspective

Welding is a quality-critical operation — the process, consumables, parameters, and thermal control are all specified for a reason.

Red Flag: E7018 electrodes left out overnight — low-hydrogen status lost

A welder removes a fresh can of E7018 electrodes from the rod oven at end of shift and leaves them on the bench. Next morning they are used without reconditioning. E7018 flux absorbs atmospheric moisture within hours — the 'low-hydrogen' designation is meaningless; hydrogen content in the deposited weld metal is now equivalent to a standard electrode. On a high-CE structural steel this is a recipe for hydrogen-induced cold cracking in the HAZ, possibly hours after the weld passes visual inspection. Reject the weld lot, recondition or discard the exposed electrodes, and document the non-conformance.

Caution: Preheat verified 30 minutes before welding — joint has since cooled

A welder preheats a thick plate joint to 150°C, confirmed by Tempilstik. Welding is delayed by 30 minutes while other work is completed. The joint has cooled to ambient. The technician must require re-preheat and re-verification immediately before the arc is struck — per AWS D1.1 the temperature must be present at the joint at the time of welding. A preheat reading taken earlier is not valid.

Good Practice: Using the weld symbol to verify the completed weld

A drawing calls for a 6 mm fillet weld, 50 mm long at 200 mm pitch, both sides of a T-joint. The technician locates the symbol on the reference line — triangle below (arrow side) and above (other side), 6 to the left, 50-200 to the right. After welding, they measure the fillet leg with a weld gauge — finds 5 mm on the arrow side. This is undersize; an NCR is raised. Reading the symbol correctly is the foundation of all weld inspection.

Core Question Answers

What is welding and how does it create a metallurgical bond?

A heat source melts the base metal edges and filler metal to form a weld pool — as it solidifies it creates a continuous metallic connection. The joint is a continuation of the metal itself, not glued. Microstructure and mechanical properties are determined by the temperatures reached and the heating and cooling rates — which is why controlling the thermal cycle is the foundation of welding quality.

What are the main arc welding processes and when is each used?

SMAW (stick): coated electrode, slag protection, portable and tolerant of wind/contamination — site work and maintenance. GMAW (MIG): continuous wire, shielding gas, high deposition rate, automatable — manufacturing and fabrication. GTAW (TIG): non-consumable tungsten, highest quality, all materials — root passes, stainless, aluminum, titanium. FCAW: flux-cored tubular wire, highest deposition rate, tolerant of mill scale — heavy structural fabrication.

How do filler metals work and how are they selected?

Filler melts into the weld pool and determines the as-welded composition, microstructure, and properties. Selected to match or overmatch base metal tensile strength. AWS classification: prefix (E/ER) + tensile strength + position + flux/gas type. E7018 = low-hydrogen SMAW electrode, 70 ksi, all positions. ER308L = low-carbon austenitic stainless GTAW/GMAW wire for 304L/316L. ER4043 = Al-Si filler for 6xxx aluminum.

What are preheat and interpass temperature and why are they quality-critical?

Preheat is the minimum base metal temperature before welding — slows cooling to prevent martensite formation and allow hydrogen to diffuse out, preventing HIC. Required preheat determined by $CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$. Interpass minimum maintains preheat benefit between passes; interpass maximum (250–300°C carbon steel, 150°C stainless) prevents grain coarsening and sensitization. Verified by Tempilstik or pyrometer immediately before welding.

How do you read a basic weld symbol?

Reference line is the horizontal baseline. Arrow points to the joint — symbols below the line apply to the arrow side, symbols above apply to the other side. Weld type symbol sits on the reference line. Number left of symbol = weld size (leg for fillets, depth for grooves). Number right = length and pitch. Circle at arrow junction = weld all around; flag = field weld. Tail specifies WPS number or process. Reading a symbol correctly is a non-negotiable inspection skill.

Key Takeaways — Week 14: Welding Processes

1

The four main arc welding processes — SMAW, GMAW, GTAW, and FCAW — each occupy a distinct niche defined by portability, productivity, quality, and base metal suitability; a technician must know not just what each process is but why it was specified for a given application and what quality risks are associated with it.

2

Preheat and interpass temperature are not optional comfort measures — they are mathematically derived quality requirements based on the carbon equivalent of the steel and the hydrogen level of the process; failing to achieve or maintain the specified temperature is a non-conformance that must be documented and addressed before welding continues.

3

Reading weld symbols is a non-negotiable literacy skill for any technician working in fabrication or inspection — a symbol that is misread means the wrong weld is made, and a wrong weld that passes inspection because the inspector did not check against the symbol is a latent defect waiting to cause a failure.

Resources & What's Next

Week 14 Resources

Kalpakjian Ch. 30 • Moniz Ch. 27

Fusion Welding Processes and Welding Metallurgy — primary reading

SMAW / GMAW / GTAW / FCAW Visualizers

Arc establishment, transfer modes, shielding gas selection — interactive

AWS D1.1 Structural Welding Code

Pre-qualified joints, filler metals, preheat, and inspection requirements

Weld Symbols — AWS A2.4

Complete standard for welding and NDT symbols — quick reference card

Preheat Calculator — Lincoln Electric

CE-based preheat recommendation by steel grade and section thickness

Week 15 — Weld Metallurgy & Failure

November 30, 2026 • Unit 5 continues

- Weld microstructure — weld metal, CGHAZ, FGHAZ, and their properties
- Hydrogen-induced cracking — the three preconditions and why it is delayed
- Hot cracking — solidification cracking and crater cracks
- Fracture types — ductile, brittle, and fatigue surface features
- Failure analysis methodology — preservation, documentation, root cause